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Course Name and Code: Database Development with PL/SQL (INSY 8311)

Lecturer's name: Eric Maniraguha

Individual Assignment I: SQL JOINS & Window Functions Project

Step 1: Problem Definition

- **Business Context:**

An e-commerce company in the retail industry wants to analyze sales performance across regions and customer segments to improve marketing and inventory planning.

Company: An e-commerce

Department: customers, products, sales

- **Data Challenge:**

The company lacks insights into product performance per region, monthly sales trends, and customer buying behavior. This hinders targeted marketing and inventory decisions.

- **Expected Outcome:**

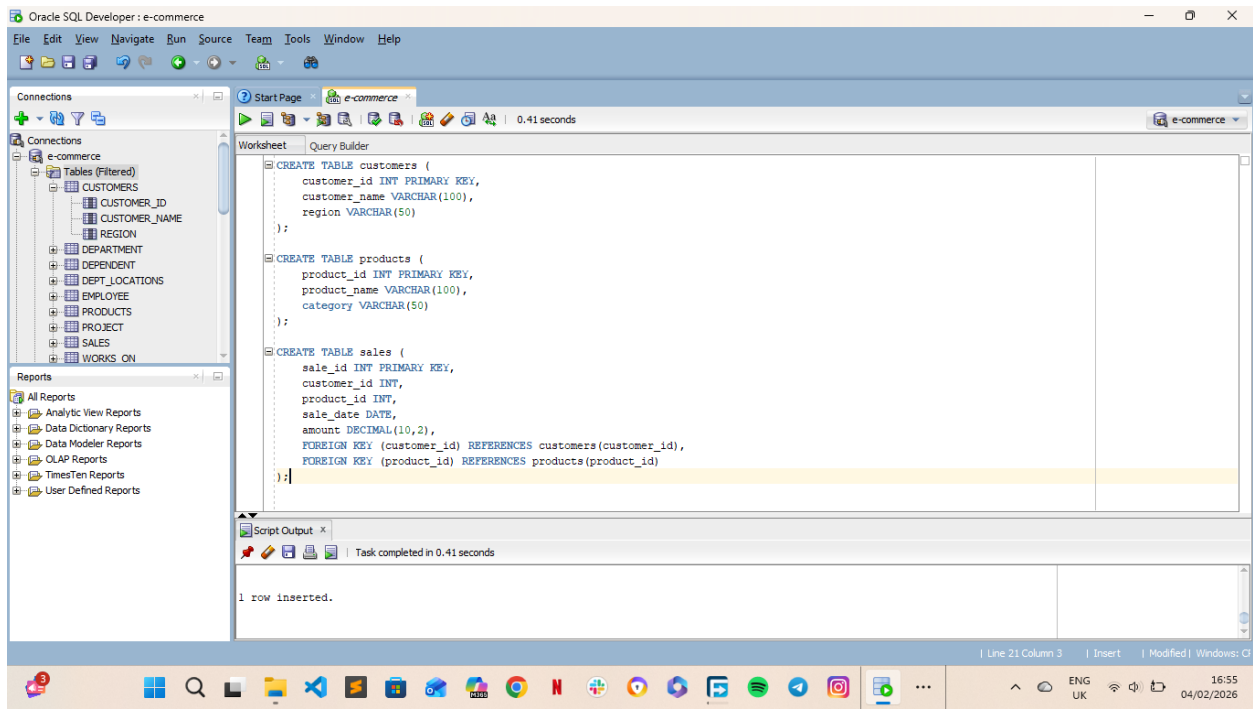
Identify top-selling products per region, track monthly sales growth, segment customers by purchase frequency, and recommend marketing actions.

Step 2: Success Criteria-

Define **5 measurable goals** linked to window functions:

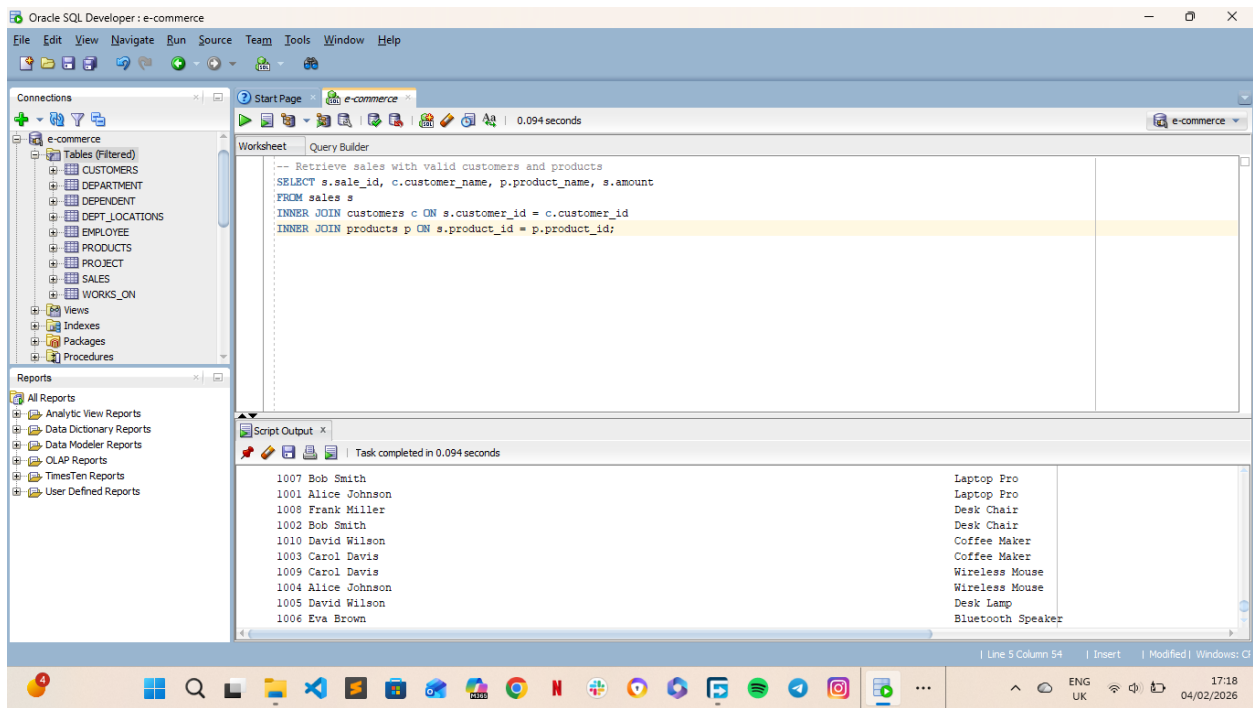
1. **Top 5 products per region** → Use RANK()
2. **Running total of monthly sales** → Use SUM() OVER() with ORDER BY
3. **Month-over-month sales growth %** → Use LAG()
4. **Customer segmentation into quartiles by total spending** → Use NTILE(4)
5. **3-month moving average of sales** → Use AVG() OVER() with ROWS BETWEEN 2 PRECEDING AND CURRENT ROW

Step 3: Database Schema Design



Step 4: Part A — SQL JOINS Implementation

1. INNER JOIN



2. LEFT JOIN

Oracle SQL Developer : e-commerce

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Connections

e-commerce

Tables (Filtered)

- CUSTOMERS
- DEPARTMENT
- DEPARTMENT
- DEPT_LOCATIONS
- EMPLOYEE
- PRODUCTS
- PROJECT
- SALES
- WORKS_ON
- Views
- Indexes
- Packages
- Procedures

Reports

- All Reports
- Analytic View Reports
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- TimesTen Reports
- User Defined Reports

Worksheet

Query Builder

```
-- Customers who never made a purchase
SELECT c.customer_id, c.customer_name
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
WHERE s.sale_id IS NULL;
```

Script Output

Task completed in 0.047 seconds

1007	Bob Smith	Laptop Pro
1001	Alice Johnson	Laptop Pro
1008	Frank Miller	Desk Chair
1002	Bob Smith	Desk Chair
1010	David Wilson	Coffee Maker
1003	Carol Davis	Coffee Maker
1009	Carol Davis	Wireless Mouse
1004	Alice Johnson	Wireless Mouse
1005	David Wilson	Desk Lamp
1006	Eva Brown	Bluetooth Speaker

17:20 04/02/2026

3. RIGHT JOIN

Oracle SQL Developer : e-commerce

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Connections

e-commerce

Tables (Filtered)

- CUSTOMERS
- DEPARTMENT
- DEPARTMENT
- DEPT_LOCATIONS
- EMPLOYEE
- PRODUCTS
- PROJECT
- SALES
- WORKS_ON
- Views
- Indexes
- Packages
- Procedures

Reports

- All Reports
- Analytic View Reports
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- TimesTen Reports
- User Defined Reports

Worksheet

Query Builder

```
-- Products with no sales
SELECT p.product_id, p.product_name
FROM sales s
RIGHT JOIN products p ON s.product_id = p.product_id
WHERE s.sale_id IS NULL;
```

Script Output

Task completed in 0.047 seconds

1008	Frank Miller	Desk Chair
1002	Bob Smith	Desk Chair
1010	David Wilson	Coffee Maker
1003	Carol Davis	Coffee Maker
1009	Carol Davis	Wireless Mouse
1004	Alice Johnson	Wireless Mouse
1005	David Wilson	Desk Lamp
1006	Eva Brown	Bluetooth Speaker

10 rows selected.

17:21 04/02/2026

4. FULL OUTER JOIN

Oracle SQL Developer : e-commerce

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Connections

- e-commerce
 - Tables (Filtered)
 - CUSTOMERS
 - DEPARTMENT
 - DEPENDENT
 - DEPT_LOCATIONS
 - EMPLOYEE
 - PRODUCTS
 - PROJECT
 - SALES
 - WORKS_ON
 - Views
 - Indexes
 - Packages
 - Procedures

Reports

- All Reports
- Analytic View Reports
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- TimesTen Reports
- User Defined Reports

Worksheet Query Builder

```
-- All customers and products, showing matches and gaps
SELECT c.customer_name, p.product_name, s.amount
FROM customers c
FULL OUTER JOIN sales s ON c.customer_id = s.customer_id
FULL OUTER JOIN products p ON s.product_id = p.product_id;
```

Script Output

Task completed in 0.064 seconds

Bob Smith	Desk Chair
David Wilson	Coffee Maker
Carol Davis	Coffee Maker
Carol Davis	Wireless Mouse
Alice Johnson	Wireless Mouse
David Wilson	Desk Lamp
Eva Brown	Bluetooth Speaker

10 rows selected.

Line 5 Column 59 | Insert | Modified | Windows: C

17:22
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6. SELF JOIN

Oracle SQL Developer : e-commerce

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Connections

- e-commerce
 - Tables (Filtered)
 - CUSTOMERS
 - DEPARTMENT
 - DEPENDENT
 - DEPT_LOCATIONS
 - EMPLOYEE
 - PRODUCTS
 - PROJECT
 - SALES
 - WORKS_ON
 - Views
 - Indexes
 - Packages
 - Procedures

Reports

- All Reports
- Analytic View Reports
- Data Dictionary Reports
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- OLAP Reports
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- User Defined Reports

Worksheet Query Builder

```
-- Compare customers in the same region
SELECT c1.customer_name AS customer1, c2.customer_name AS customer2, c1.region
FROM customers c1
JOIN customers c2 ON c1.region = c2.region AND c1.customer_id < c2.customer_id;
```

Script Output

Task completed in 0.048 seconds

CUSTOMER1	CUSTOMER2
Alice Johnson	Carol Davis
Bob Smith	Frank Miller

10 rows selected.

Line 4 Column 80 | Insert | Modified | Windows: C

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Step 5: Part B — Window Functions Implementation

1. Ranking Functions

Oracle SQL Developer : e-commerce

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Connections

Start Page e-commerce 0.063 seconds

Worksheet Query Builder

```
-- Top 5 products by revenue per region
WITH product_revenue AS (
  SELECT
    region,
    product_name,
    SUM(amount) AS revenue,
    RANK() OVER(PARTITION BY region ORDER BY SUM(amount) DESC) AS rank
  FROM sales s
  JOIN customers c ON s.customer_id = c.customer_id
  JOIN products p ON s.product_id = p.product_id
  GROUP BY region, product_name
)
SELECT * FROM product_revenue WHERE rank <= 5;
```

Script Output

Task completed in 0.063 seconds

North	Laptop Pro
North	Coffee Maker
North	Wireless Mouse
South	Laptop Pro
South	Desk Chair
West	Bluetooth Speaker

8 rows selected.

Line 13 Column 47 | Insert | Modified | Windows: C

17:26 04/02/2026

2. Aggregate Window Functions

Oracle SQL Developer : e-commerce

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Connections

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Worksheet Query Builder

```
-- Running total of sales by month
SELECT
  TRUNC(sale_date, 'MM') AS month,
  SUM(amount) AS monthly_sales,
  SUM(SUM(amount)) OVER(ORDER BY TRUNC(sale_date, 'MM')) AS running_total
FROM sales
GROUP BY TRUNC(sale_date, 'MM')
ORDER BY month;
```

Script Output

Task completed in 0.462 seconds

'Cause:
'Action:

MONTH	MONTHLY_SALES	RUNNING_TOTAL
01-JAN-24	1350	1350
01-FEB-24	1470	2820
01-MAR-24	255	3075

Line 8 Column 15 | Insert | Modified | Windows: C

17:33 04/02/2026

3. Navigation Functions

Oracle SQL Developer: e-commerce

Connections: e-commerce

Worksheet: Query Builder

```

WITH monthly AS (
  SELECT
    TRUNC(sale_date, 'MM') AS month,
    SUM(amount) AS revenue
  FROM sales
  GROUP BY TRUNC(sale_date, 'MM')
)
SELECT
  month,
  TO_CHAR(month, 'Mon YYYY') AS month_display,
  revenue,
  LAG(revenue, 1) OVER(ORDER BY month) AS prev_month,
  ROUND(((revenue - LAG(revenue, 1) OVER(ORDER BY month)) /
    NULLIF(LAG(revenue, 1) OVER(ORDER BY month), 0)) * 100, 2) AS growth_percent
FROM monthly
ORDER BY month;

```

Script Output: Task completed in 0.043 seconds

MONTH	MONTH_DI	REVENUE	PREV_MONTH	GROWTH_PERCENT
01-JAN-24	Jan 2024	1350		
01-FEB-24	Feb 2024	1470	1350	8.89
01-MAR-24	Mar 2024	255	1470	-82.65

Line 17 Column 16 | Insert | Modified | Windows: C

17:36 04/02/2026

4. Distribution Functions

Oracle SQL Developer: e-commerce

Connections: e-commerce

Worksheet: Query Builder

```

-- Customer segmentation into quartiles by total spending
WITH customer_spending AS (
  SELECT
    customer_id,
    SUM(amount) AS total_spent,
    NTILE(4) OVER(ORDER BY SUM(amount)) AS spending_quartile
  FROM sales
  GROUP BY customer_id
)
SELECT * FROM customer_spending
ORDER BY spending_quartile, total_spent DESC;

```

Script Output: Task completed in 0.046 seconds

5	120	1
3	105	1
6	150	2
4	125	2
1	1225	3
2	1350	4

6 rows selected.

Line 11 Column 46 | Insert | Modified | Windows: C

17:40 04/02/2026

Step 6: GitHub Repository

- README.md must include:
 - Business problem
 - Schema and ER diagram

- JOIN queries
- Window function queries
- Key insights
- References
- Integrity statement
- Screenshots of results

- **Step 7: Results Analysis**

- **Descriptive:** Top products are in Region A, monthly sales peaked in December.
Diagnostic: High sales in December due to holiday season; low sales in Region B due to poor marketing.
Prescriptive: Increase marketing in Region B, launch promotions for low-selling products, re-engage inactive customers.

Step 8: References

Add:

1. Oracle SQL Documentation
2. Academic paper on customer segmentation
3. Business analytics blog on sales trends

And the integrity statement:

"All sources were properly cited. Implementations and analysis represent original work. No AI-generated content was copied without attribution or adaptation."